

one image, one set of logits

cat
z = +2.4 true

dog
z = -0.5

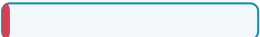
outdoors
z = +2.1 true

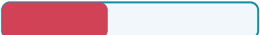
the photograph is a cat, and it is outdoors — both

softmax

$$e^{z_i} / \sum_j e^{z_j}$$

cat  0.557

dog  0.031

outdoors  0.413

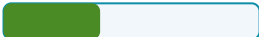
sum 1.000

always 1 — the classes share one budget

sigmoid, per class

$$1 / (1 + e^{-z})$$

cat  0.917

dog  0.378

outdoors  0.891

sum 2.185

not 1, and nothing says it should be